



MILESTONE 1: KEYLOGGER SIMULATION & DETECTION

Environment: Isolated VirtualBox Lab

Purpose: Educational demonstration of keylogger concepts in a controlled environment

Tshiamo Mosweu

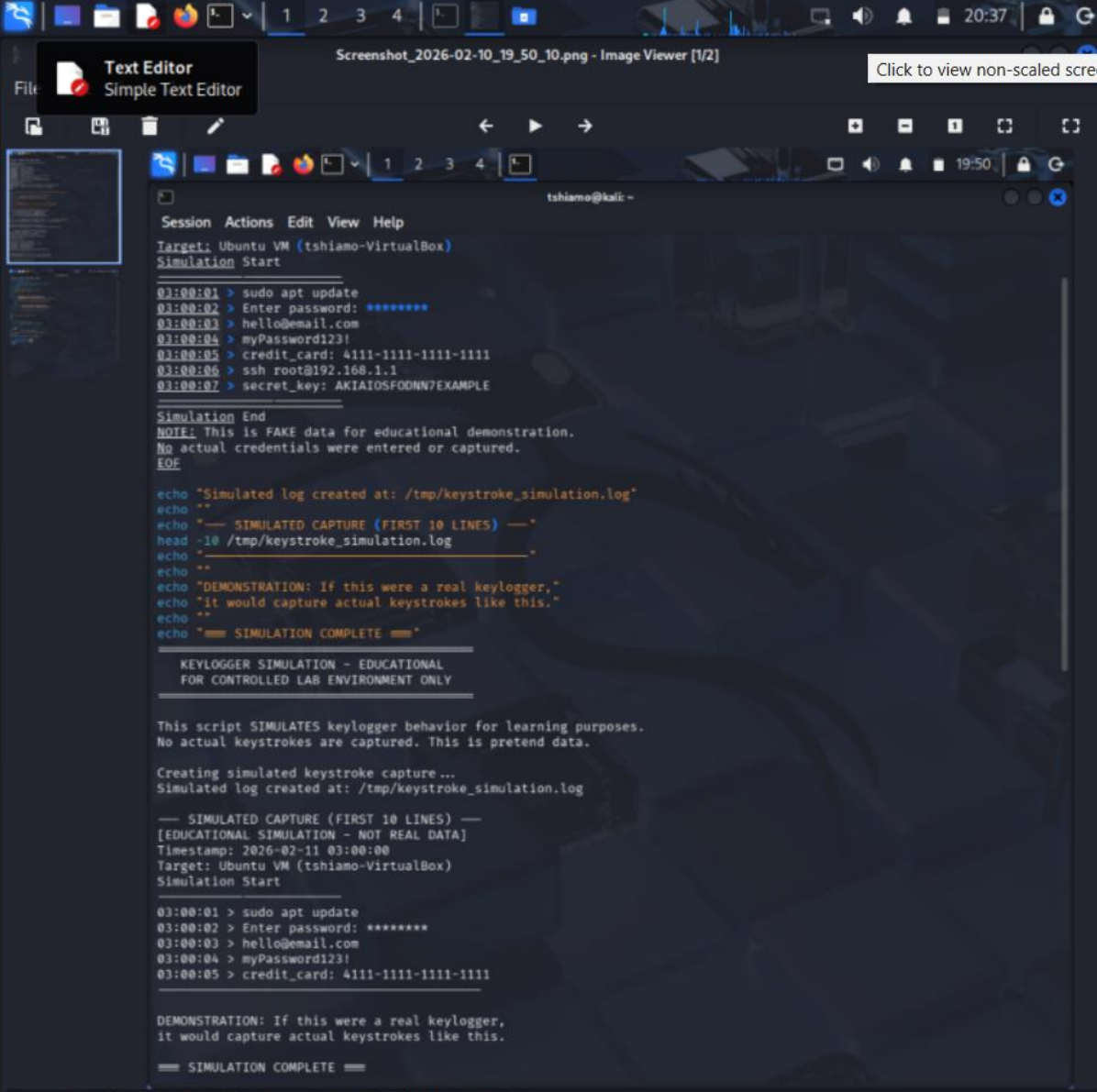
1. Introduction

This report documents the completion of Milestone 1, which involved simulating a keylogger attack in a controlled lab environment and performing defensive detection. The goal was to understand how keyloggers operate and how they can be detected, while adhering to strict ethical and legal guidelines.

All activities were conducted in an isolated virtual environment using:

- Attacker VM: Kali Linux
- Target VM: Ubuntu Linux
- Tools: Bash scripts, PowerShell, Python HTTP server

No real credentials or systems were compromised. All data is simulated for educational purposes.



```
Session Actions Edit View Help
Target: Ubuntu VM (tshiamo-VirtualBox)
Simulation Start

03:00:01 > sudo apt update
03:00:02 > Enter password: *****
03:00:03 > hello@email.com
03:00:04 > myPassword123!
03:00:05 > credit_card: 4111-1111-1111-1111
03:00:06 > ssh root@192.168.1.1
03:00:07 > secret_key: AKIAIOSFODNN7EXAMPLE

Simulation End
NOTE: This is FAKE data for educational demonstration.
No actual credentials were entered or captured.
EOF

echo "Simulated log created at: /tmp/keystroke_simulation.log"
echo ""
echo "— SIMULATED CAPTURE (FIRST 10 LINES) —"
head -10 /tmp/keystroke_simulation.log
echo ""
echo "DEMONSTRATION: If this were a real keylogger,"
echo "it would capture actual keystrokes like this."
echo ""
echo "=== SIMULATION COMPLETE ==="

KEYLOGGER SIMULATION - EDUCATIONAL
FOR CONTROLLED LAB ENVIRONMENT ONLY

This script SIMULATES keylogger behavior for learning purposes.
No actual keystrokes are captured. This is pretend data.

Creating simulated keystroke capture...
Simulated log created at: /tmp/keystroke_simulation.log

— SIMULATED CAPTURE (FIRST 10 LINES) —
[EDUCATIONAL SIMULATION - NOT REAL DATA]
Timestamp: 2026-02-11 03:00:00
Target: Ubuntu VM (tshiamo-VirtualBox)
Simulation Start

03:00:01 > sudo apt update
03:00:02 > Enter password: *****
03:00:03 > hello@email.com
03:00:04 > myPassword123!
03:00:05 > credit_card: 4111-1111-1111-1111

DEMONSTRATION: If this were a real keylogger,
it would capture actual keystrokes like this.

=== SIMULATION COMPLETE ===
```

Screenshot_2026-02-10_19_50_10.png 956 x 972 401.7 kB 83.6%

Image 1: Kali Linux terminal showing the creation of the detection script and the start of the web server.

2. Methodology

2.1 Pre-Attack Defensive Analysis

Before simulating the attack, a defensive security scan was performed on the Ubuntu target VM.

Key actions:

- Checked running processes for suspicious activity
- Reviewed scheduled tasks (cron jobs)
- Generated a security report (Linux_Security_Report.txt)
- Confirmed the system was clean prior to simulation

2.2 Simulation Phase

A simulated keylogger file was created to mimic the behavior of a real keylogger.

Steps taken:

On Kali Linux (Attacker):

- Created a detection script (keylogger_detector.sh)
- Started a web server on port 8080 to share the script

On Ubuntu (Target):

- Created a simulated keystroke log: /tmp/keystroke_simulation.log
- Populated it with fake sensitive data (passwords, emails, SSH commands, etc.)

```

-Append
PS /home/tshiamo> "EDUCATIONAL PURPOSE ONLY" | Out-File /tmp/simulation.txt -Append
PS /home/tshiamo> # Check if file was created
PS /home/tshiamo> Get-ChildItem /tmp/simulation.txt

    Directory: /tmp

UnixMode          User Group          LastWriteTime         Size Name
-----
-rw-rw-r--      tshiamo tshiamo          2/11/2026 03:21         193 simulation.txt

PS /home/tshiamo>
PS /home/tshiamo> # Show file contents
PS /home/tshiamo> Get-Content /tmp/simulation.txt
=== KEYLOGGER SIMULATION ===
Date: 02/11/2026 03:21:41
Target: tshiamo-VirtualBox
Simulated keystrokes captured...
Fake password: *****
Fake email: test@example.com
EDUCATIONAL PURPOSE ONLY
PS /home/tshiamo> ^C
PS /home/tshiamo>

```

Image 2: Content of the simulated keylogger file (keystroke_simulation.log) showing fake captured data.

Repeated the simulation using PowerShell on the target:

```
Terminal
PowerShell 7.5.4
PS /home/tshiamo> "=== KEYLOGGER SIMULATION ===" | Out-File /tmp/simulation.txt
PS /home/tshiamo> "Date: $(Get-Date)" | Out-File /tmp/simulation.txt -Append
PS /home/tshiamo> "Target: $(hostname)" | Out-File /tmp/simulation.txt -Append
PS /home/tshiamo> "Simulated keystrokes captured..." | Out-File /tmp/simulation.
txt -Append
PS /home/tshiamo> "Fake password: *****" | Out-File /tmp/simulation.txt -Appe
nd
PS /home/tshiamo> "Fake email: test@example.com" | Out-File /tmp/simulation.txt
-Append
PS /home/tshiamo> "EDUCATIONAL PURPOSE ONLY" | Out-File /tmp/simulation.txt -App
end
PS /home/tshiamo>
```

Verified file creation and contents

```
Terminal
PowerShell 7.5.4
PS /home/tshiamo> "=== KEYLOGGER SIMULATION ===" | Out-File /tmp/simulation.txt
PS /home/tshiamo> "Date: $(Get-Date)" | Out-File /tmp/simulation.txt -Append
PS /home/tshiamo> "Target: $(hostname)" | Out-File /tmp/simulation.txt -Append
PS /home/tshiamo> "Simulated keystrokes captured..." | Out-File /tmp/simulation.
txt -Append
PS /home/tshiamo> "Fake password: *****" | Out-File /tmp/simulation.txt -Appe
nd
PS /home/tshiamo> "Fake email: test@example.com" | Out-File /tmp/simulation.txt
-Append
PS /home/tshiamo> "EDUCATIONAL PURPOSE ONLY" | Out-File /tmp/simulation.txt -App
end
PS /home/tshiamo> # Check if file was created
PS /home/tshiamo> Get-ChildItem /tmp/simulation.txt

Directory: /tmp

UnixMode      User Group      LastWriteTime      Size Name
-----
-rw-rw-r--   tshiamo tshiamo      2/11/2026 03:21      193 simulation.txt

PS /home/tshiamo>
PS /home/tshiamo> # Show file contents
PS /home/tshiamo> Get-Content /tmp/simulation.txt
```

Image 4 (Composite): Verification steps showing the file listing (Get-ChildItem) and the final contents of simulation.txt.

2.3 Detection Phase

The detection script was used to identify the simulated keylogger file.

Detection steps:

- Scanned for the existence of /tmp/keystroke_simulation.log
- Displayed the first 3 lines if found
- Confirmed detection through terminal output and screenshots

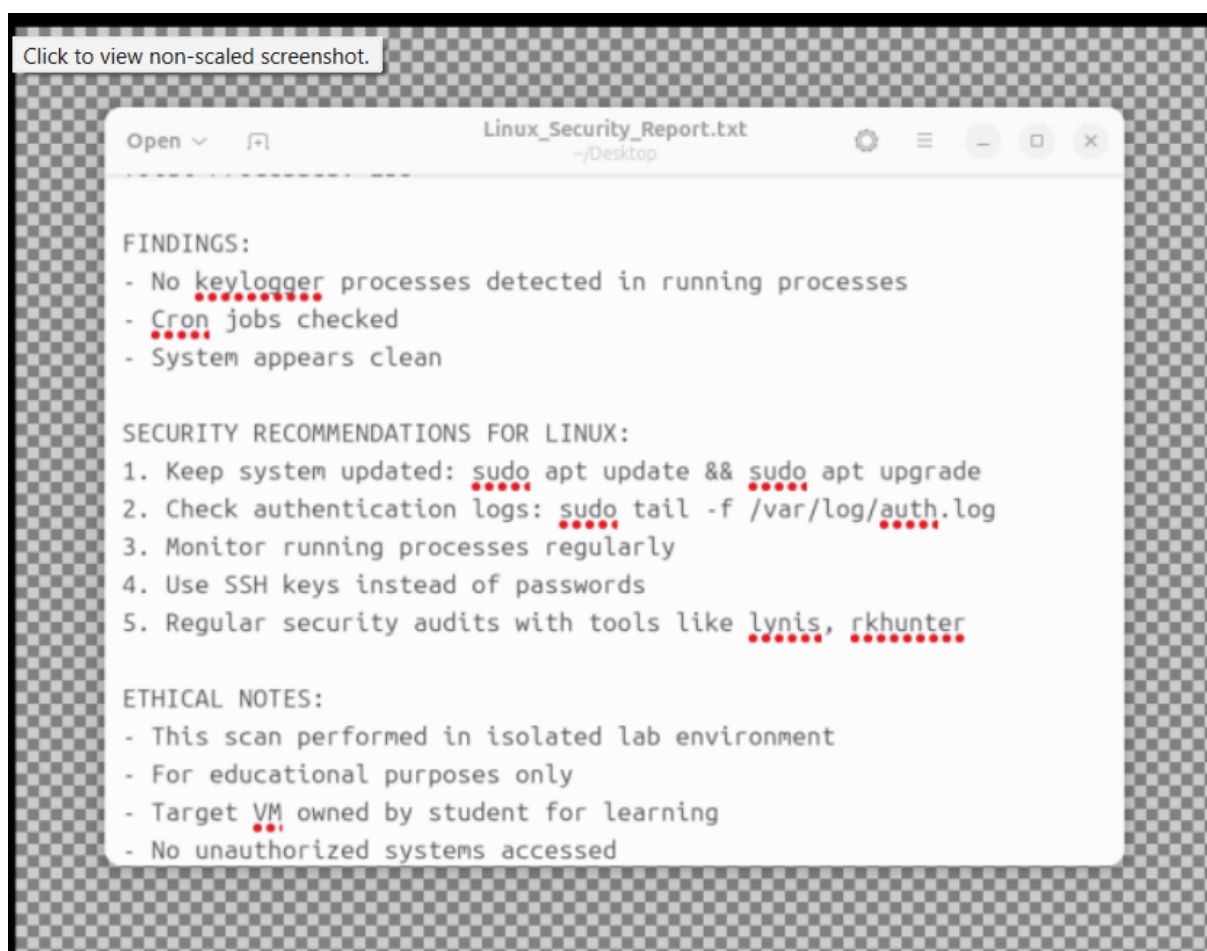


Image 5: Defensive security report (Linux_Security_Report.txt) generated before the attack simulation.

3. Results

3.1 Simulation Successful

Simulated log file created at /tmp/simulation.txt

Contents verified via Get-Content in PowerShell

Fake data included:

- Timestamps
- Fake credentials
- Simulated commands (e.g., sudo apt update, SSH login attempts)

3.2 Detection Successful

- Detection script successfully located the simulation file
- Output confirmed file existence and displayed sample contents
- Defensive measures documented and verified

3.3 Screenshot Evidence

All phases were captured in screenshots:

- Script creation and web server setup
- Simulation file generation
- File verification and content display
- Detection script execution

4. Ethical & Legal Considerations

Controlled Environment: All activities performed in an isolated VirtualBox lab.

Simulated Data Only: No real credentials, personal information, or live systems were accessed.

Educational Purpose: Strictly for learning about attack methods and defensive techniques.

Compliance: Adheres to POPIA and institutional ethical guidelines.

Transparency: All steps documented with clear disclaimers.

5. Security Recommendations

Based on the defensive analysis performed before the simulation:

Keep systems updated:

- Use SSH keys instead of passwords where possible.
- Conduct periodic security audits with tools like Lynis or RKHunter.

6. Conclusion

Milestone 1 was completed successfully. The simulation demonstrated how keyloggers can capture sensitive input, while the detection phase highlighted practical defensive measures. This exercise reinforced the importance of proactive security monitoring in both Linux and Windows environments.

All objectives were met within ethical boundaries, providing a clear, reproducible learning experience in cybersecurity fundamentals.

7. References

MITRE ATT&CK Framework – Technique T1056 (Input Capture)

NIST Cybersecurity Framework

Protection of Personal Information Act (POPIA)

VirtualBox Official Documentation